

1. Overview & Core Features

The **File Repository Service** is a lightweight, secure web application powered by a Python/Flask backend and an intuitive single-page interface. This document and media management application was designed for intuitive local and network file storage. Key operational capabilities include:

- **Interactive Directory Navigation:** Breadcrumb top-navigation and parent-directory quick jumps.
- **Drag-and-Drop File & Directory Uploads:** Support for multi-file and complete directory tree uploads with recursive folder scanning.
- **Automatic Conflict Handling:** Built-in conflict resolution (Merge/Overwrite, Suffix Rename, or Cancel) for duplicate files or folders.
- **Batch Processing & Progress Tracking:** Real-time upload percentage tracking with automatic batching (5 MB batches) and retries for network resilience.
- **Repository Structure Mapping:** Interactive visual folder tree rendering file counts across the repository.
- **Global Search with Direct Navigation:** Fast map-indexed search that directly opens the parent folder of matching files or presents multi-match jump links.
- **Rich In-Browser File Previews:** Direct viewing for standard images, high-resolution TIFF images (via UTIF.js), PDFs, HTML/code/text documents, and audio/video media without requiring external downloads.
- **Soft Deletion Safety:** Deleted files and directories are automatically safely moved into a dedicated `deleted_files` archive directory.
- **Storage Directories:**
 - `shared_files/`: Root directory for all active user files and folders.
 - `deleted_files/`: Safety archive directory where deleted items are preserved.
- **Security & Path Validation:** All user requests undergo strict path traversal verification to ensure operations remain safely contained within the repository bounds.
- **Server Upload Thresholds:** Enforces a maximum batch request limit of **500 MB** per batch submission, while frontend JavaScript chunks files into smaller 5 MB payloads for optimal transfer reliability.

2. Directory Navigation & Folder Management

Directory Navigation

- **Breadcrumb Navigation:** Click **Root** or any intermediate folder segment in the top navigation bar to jump directly to that directory level.
- **Folder Browsing:** Click any folder link marked with  to enter it.
- **Parent Directory Jump:** Click  **.. (Parent Directory)** at the top of the item table to step back up one folder level.

Creating New Folders

1. Open the directory where the new subfolder should be created.
2. Type the folder name into the **New Folder Name** field in the action bar.
3. Click  **Create Folder**. The system creates the directory immediately and regenerates the repository structure map.

3. File & Folder Uploads

Upload Methods

- **Drag-and-Drop:** Drag one or more files—or entire folder structures from your computer—directly into the dashed **Drag & Drop** drop zone.
- **Select Files Button:** Click  **Select Files** to pick individual or multiple files via your system file picker.
- **Select Folder Button:** Click  **Select Folder** to select a full folder hierarchy. Web browser security limits selection to one folder. For multiple folder uploads use the the Drag-and-Drop method.

Stream Processing & Performance

- **Single File Limit:** Files over 100 MB trigger a confirmation prompt to skip or proceed.
- **Chunked Uploads:** Files are transmitted in 5 MB frontend batches and streamed directly to disk in **64 KB chunks**, keeping memory usage minimal.
- **Server Request Ceiling:** The total batch payload limit supported by the server is **500 MB**.

Conflict Handling Options

If an uploaded file or folder path already exists, a conflict dialog prompts you to choose:

1. **Merge folder / overwrite files (Orange):** Preserves existing folder structures, adds new files, and replaces matching files. (Note: A file cannot overwrite a folder, nor vice-versa).
2. **Rename with suffix (Blue):** Appends a numerical index (e.g., (1)) to file or folder names to avoid overwriting existing data.
3. **Cancel upload (Grey):** Aborts the upload operation.

4. File Operations: Download, Preview & Delete

Downloading Items

- **Single Files:** Click  **Download** next to any file row.
- **Folders:** Click  **Download Zip** next to any folder row to download the entire directory packaged as a compressed .zip archive.

In-Browser File Previews

Click  **Preview** next to a supported file to open the inline modal player/viewer:

Supported Media Type	Extension / Format	Preview Behavior
Standard Images	.jpg, .jpeg, .png, .gif, .webp, .svg, .bmp	Direct rendering in preview pane.
TIFF Images	.tif, .tiff	Decoded on-the-fly in browser via UTIF.js onto an HTML canvas.
Video Files	.mp4, .webm, .ogg, .mkv	Embedded HTML5 video player with playback controls.
Audio Files	.mp3, .wav, .ogg, .m4a, .flac	Embedded HTML5 audio player with playback controls.
PDF Documents	.pdf	Embedded interactive PDF viewer iframe.
Text & Code	.txt, .md, .json, .csv, .py, .html, .css, .js, .xml, .log, .ini, .cfg	Dark-mode monospaced code view with HTML character escaping.

Notes:

- For unsupported file formats, use the green **Download** button to view the file locally on your computer.
- When viewing PDFs, the file will be downloaded and the opened in your browser's PDF viewer. The file will remain in your Downloads location since the application cannot remove files on your device.

Deleting Items (Soft Delete Safety)

When you click **Delete** and confirm:

1. The file or folder is moved from `shared_files/` to `deleted_files/` while maintaining its original subfolder structure.
2. **Timestamp Conflict Safety:** If an item with the exact same name already exists in `deleted_files/`, the server automatically appends a microsecond timestamp (e.g., `filename.deleted-202610071347000000-1.ext`) to ensure previous backups are never overwritten.

5. Repository Mapping & Global Search

Generating and Viewing the Repository Structure Map

1. **Regenerate Map:** Click  **(Re)Generate Map** in the top-right toolbar to force the server to index the latest folder and file layout across the entire storage repository.
2. **View Structure:** Click  **View Repository Map** to display an interactive, indented tree view showing every folder, subfolder, and the count of contained files. Clicking any folder in the map modal jumps directly into that directory.

Global Search

1. Enter a file name or keyword into the **Search for files...** box and click **Search Map**.
2. **Matching Logic:** Case-insensitive substring match against both file and folder names across the entire repository.
3. **Single Match Result:** The browser immediately navigates directly to the parent directory containing the matching file.
4. **Multiple Matches Result:** A **Multiple Matches Found** dialog displays direct jump links for every matching item path.

6. Troubleshooting & FAQs

- **Why did my upload stop or fail?** Ensure individual files do not exceed 100 MB. If a network timeout occurs during multi-file batch transfers, the system automatically attempts retries. Check that your internet or local connection remains active during large folder transfers.
- **What happens if an upload fails halfway?** Transfers run in auto-retrying batches. If an error occurs, the server reports the status, and retries occur automatically before failing gracefully. Refresh the browser tab to see the current state.
- **What is Error 413 ?** HTTP error 413, commonly referred to as "Payload Too Large," occurs when a server refuses to process a request because the size of the data being sent exceeds the server's allowed limits. This can happen when trying to upload a folder hierarchy with hundreds or thousands of items in it. You will have to upload the data in multiple steps
- **Why can't I preview my TIFF image?** Very large or uncommon TIFF compression modes may fail browser canvas decoding. Use the **Download** link to view the file in your system's native image viewer.
- **Where do deleted files go?** Items deleted via the user interface are not permanently destroyed immediately; they are moved into the system's deleted_files directory for recovery or administrator cleanup.
- **My new folder/file isn't appearing in Search or Map.** Search relies on the repository index. Click  **(Re)Generate Map** in the upper toolbar to update the index with your recent modifications.